Self-destructive feeding and nonderived environment blocking in Parallel OT: an underspecification account

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Abstract Self-destructive feeding (SDF) is a type of overapplication opacity in which, in rule-based terms, an earlier rule creates the environment for a later rule that in turn removes part of the environment that motivated the earlier one. This paper re-examines four cases of SDF (two in Turkish, one in Javanese and one in Japanese) and shows that they can be accounted for in Parallel Optimality Theory. After introducing the four cases, the paper presents the two properties they share. First, at least one of the processes involved in each interaction is restricted to morpheme junctures and particular morphemes. Second, they resolve consonant sequences at stem–suffix junctures by realising a suffix-initial segment in required contexts. I propose that the alternating segments are underspecified for their association to the skeletal tier, whereas non-alternating segments are fully-specified. In Parallel Optimality Theory, faithfulness to association in general and stronger faithfulness in certain input contexts or positions then account for the lack of morpheme-internal alternations and the SDF outputs in a single input–output evaluation. The three SDF cases involving consonants follow from the rankings needed for the simple alternations alone, and the vowel case requires only one additional ranking. Therefore, SDF involving alternations with ∅ can be derived in Parallel Optimality Theory once the underlying representations of segments are more appropriately understood.

1 Introduction

Opacity, as defined in (1), has received sustained attention from generative phonologists because it bears directly on the architecture of the phonological grammar.

(1) Opacity (Kiparsky 1973: 79)

A phonological rule ℙ of the form A → B / C _ D is opaque if there are surface structures with either of the following characteristics:

a. instances of A in the environment C _ D (i.e. *underapplication*), or
b. instances of B derived by ℙ that occur in environments other than C _ D (i.e. *overapplication*).

In rule-based phonology (Chomsky & Halle 1968, *a.o.*), opacity results from extrinsic rule ordering: in the classic taxonomy, counterfeeding leads to underapplication and counterbleeding to overapplication (McCarthy 2007a; b; Baković 2011). Parallel Optimality Theory (OT; Prince & Smolensky 1993) encounters more difficulty with opacity, because its surface-orientedness favours transparent candidates which usually incur fewer fatal violations, and it has no intermediate representation to justify the changes required by certain rules.

Beyond the classic taxonomy, *self-destructive feeding* (SDF) is another source of overapplication. It arises when "an earlier rule feeds a later rule that in turn crucially changes the string such that the earlier rule's application is no longer justified" (Baković 2011: 59). For example, in Turkish, a high vowel alternates with ∅ at the beginning of some suffixes. The second person singular possessive is [-n] after a vowel ([annen] 'your mother') but [-in] after a consonant ([ipin] 'your rope'). On a common rule-based analysis, Epenthesis inserts a vowel to break up an illicit word-final consonant cluster. In addition, Velar Deletion removes velar stops at the end of many polysyllabic nouns when a vowel-initial suffix is attached (Lees 1961; Underhill 1976; Sezer 1981), as in [bebek] 'baby' vs. [bebei] 'baby (ACC)'. So, in /bebek+n/, Epenthesis creates the intervocalic environment for Velar Deletion to apply, and Velar Deletion removes the /k/ that was part of the structural description of Epenthesis. The surface form [bebein] 'your baby' therefore exhibits overapplication because an inserted vowel appears in the absence of a word-final cluster on the surface (2).

(2) SDF with vowels in Turkish (Sprouse 1997; Baković 2007)

	UR	a. /bebek+n/	b. /ip+n/	c. /bebek+i/
Epenthesis:[1]	∅ → i / C _ C#	i	i	
Velar Deletion:[2]	k/g → ∅ / V_V	∅		∅
	SF	[bebein]	[ipin]	[bebei]
		'your baby'	'your rope'	'baby (ACC)'

[1] Whether this vowel is inserted or present underlyingly is debated. Lees (1961: 40) posits an underlying suffix-initial vowel that is deleted after vowels, whereas Hankamer (2011) argues that it is epenthetic (see also Yavaş 1980). The derivation in (2) follows Baković (2007), who treats the alternation as epenthesis. The analysis proposed here represents the vowel underlyingly as part of the suffix. Section 4.2 discusses this choice.

[2] Velar Deletion has lexical exceptions and further conditions (e.g. it does not apply to verb roots, to monosyllabic stems or after long vowels; Inkelas 2011). These are discussed in Section 3.1. Özçelik (2024: 74) argued that the segments being deleted in these positions are actually velar glides [ɯ] instead of stops. I will keep using /k/ and refer to the deleted segments as 'velars'. The manner of the velar consonant being

The interaction poses a problem for Parallel OT because the attested output [bebein] departs from the input in two ways (an inserted vowel and a deleted consonant), whereas its competitor [beben] avoids both the cluster and the intervocalic velar with only one deletion (Baković 2007: 229). Previous constraint-based analyses of SDF have largely fallen into two categories: (i) those that allow an intermediate stage or corresponding candidates in a derivation, such as Sympathy (Lee 1999) and OT with Candidate Chains (OT-CC; Lee 2007), and (ii) those that enrich output representations so that a segment is present for some constraints but absent for others, such as turbidity (Baković 2007). But these analyses either have to forfeit a key element (i.e. parallelism) of OT, or make ad-hoc adjustments to surface representations.

This paper takes a different approach to the treatment of SDF, based on the two properties shared by all four reported cases of SDF examined here: two in Turkish, one in Javanese and one in Japanese. First, at least one of the processes involved in each interaction is confined to morpheme junctures, and even at junctures it is restricted to particular morphemes. Second, they avoid consonant clusters at stem–suffix junctures in the same way, namely by selecting the consonant-initial form to follow vowel-final stems and the vowel-initial form to follow consonant-final stems. I argue that both properties can be explained if the alternating segments are *underspecified*: they are present in the underlying representation as melodic units, but lack an association to the skeletal tier (Kiparsky 1993; Rasin 2025). Non-alternating segments, including lexical exceptions, are fully-specified. By requiring faithfulness to association in general and stronger faithfulness in certain input contexts or positions, the grammar can derive the simple alternations in these languages and explain why the processes are confined to morpheme junctures in Parallel OT.

Consequently, the SDF outputs also naturally fall out from the same constraint rankings, with only one additional ranking for the Turkish SDF involving vowels. SDF is thus not a phenomenon that a grammar needs to model in the form of rule interaction: it is what happens when two underspecified segments meet at a juncture. This reanalysis shows that the cases usually described as SDF do not require serial derivations or intermediate stages once the representation of the alternating segments is taken into account.

The paper is arranged as follows. Section 2 presents the four cases with their full paradigms and explains why they are problematic for Parallel OT. Section 3 identifies the two properties they share. Section 4 develops the analysis for Turkish, motivating each ranking in turn, and extends it to Javanese and Japanese. Section 5 compares it with previous analyses of SDF and of NDEB in Parallel OT. Section 6 discusses the architectural commitments of the proposal: Richness of the Base, the expressivity of contextual faithfulness, the factorial typology, and the scope of the account. Section 7 concludes. The Javanese and Japanese tableaux are given in Appendix A, and Appendix B presents the computational verification of the rankings.

2 Four cases of SDF

This paper focuses on four cases of SDF: two in Turkish, one in Javanese and one in Japanese. Other cases that can be analysed as SDF also exist, but they are set aside here because they differ significantly from the four cases below that can only result from SDF.[3]

removed does not affect the argument, since the change involved in this process is deletion rather than a change of feature.

[3] The concealed free rides of Cibaeño Spanish (Golibart 1976; Guitart 1981) involve SDF on focus, where the later rule removes the focus of the earlier one, rather than on environment, where the later rule destroys

2.1 Turkish

Two SDF cases have been reported for Turkish. I refer to them as *vowel-SDF*, in which the alternating segment of the suffix is a vowel (2), and *consonant-SDF*, in which it is a consonant (3). Both cases involve stem-final velars and suffix-initial segments alternating with ∅. Table 1 gives the relevant paradigms.[4] The stem-final velars surface in the bare form and before the plural suffix *-lAr*, whose suffix-initial segment never alternates. The velar is deleted when a vowel-initial suffix such as the accusative *-(y)I* is attached.[5] The second and third person singular possessives *-(I)n* and *-(s)I* surface as vowel-initial only after consonant-final stems and consonant-initial only after vowel-final stems.

	Bare	PL *-lAr*	ACC *-(y)I*	2SG.POSS *-(I)n*	3SG.POSS *-(s)I*
a. bebek 'baby'	[bebek]	[bebekler]	[bebei]	[bebein]	[bebei]
b. ayak 'foot'	[ajak]	[ajaklar]	[ajaɯ]	[ajaɯn]	[ajaɯ]
c. ip 'rope'	[ip]	[ipler]	[ipi]	[ipin]	[ipi]
d. çan 'bell'	[ʧan]	[ʧanlar]	[ʧanɯ]	[ʧanɯn]	[ʧanɯ]
e. anne 'mother'	[anne]	[anneler]	[anneji]	[annen]	[annesi]
f. şişe 'bottle'	[ʃiʃe]	[ʃiʃeler]	[ʃiʃeji]	[ʃiʃen]	[ʃiʃesi]

Table 1: Turkish paradigms

Vowel-SDF was illustrated in (2). Consonant-SDF is shown in (3). Elision deletes /s/ or /j/ after a consonant ([ʧanɯ] vs. [ʃiʃesi]), and Velar Deletion removes the stem-final velar once it has become intervocalic. In /ajak+sɯ/, Elision creates the environment for Velar Deletion to apply, but Velar Deletion destroys a crucial part of the structural description of Elision.

(3) SDF with consonants in Turkish (Baković 2007, citing Kenstowicz & Kisseberth 1979)[6]

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the necessary context for the earlier rule to apply, as in the four cases here (Baković 2011). They can be analysed as cross-derivational feeding (Baković 2005; see also Baković 2007: 247–251). The Ramsau German cases analysed as SDF by Noelliste (2017) involve Vowel Nasalisation and Nasal Deletion, and Nasal Place Assimilation and Voiced Stop Deletion. The SDF analysis of these two cases depends on the second rule referring to a feature supplied by the first. However, Nasal Deletion does not need to refer to the nasality of the preceding vowel created by the first rule; the same applies to Voiced Stop Deletion. So, these two interactions may be analysed as counterbleeding instead of SDF. These cases are also open to a coalescence analysis (Zaleska 2020; see also Lamont 2025 for a coalescence account for counterbleeding). This alternative analytical possibility makes Cibaeño Spanish and Ramsau German differ from the four cases discussed here, which can only be described as SDF.

[4] Unless otherwise stated, the Turkish data are from the Turkish Electronic Living Lexicon (TELL: http://linguistics.berkeley.edu/TELL/). Vowels are transcribed as the context requires under vowel harmony.

[5] I choose to analyse the accusative *-(y)I* as underlyingly vowel-initial and as interacting in the way shown in (2). But, the [j] in *-(y)I* could also be analysed like the [s] of *-(s)I*. If so, forms such as [bebei] would be derived in the same way as consonant-SDF cases (Section 4.5).

[6] Baković (2007) represents the stem-final velars in 'his/her foot' with voiced /g/ and so does Hauser (2016) (see Section 5.1.1). Özçelik (2024: 83–84) shows that Turkish stem-final obstruents are voiceless in bare form and when followed by consonant-initial suffixes, and that they are voiced when followed by vowel-initial suffixes. I choose the voiceless /k/ following TELL; this choice does not affect the proposed analysis, which concerns deletion rather than voicing change.

	UR	a. /ajak+sɯ/	b. /ʧan+sɯ/	c. /bebek+i/
Elision:	s/j → ∅ / C_	∅	∅	
Velar Deletion:	k/g → ∅ / V_V	∅		∅
	SF	[ajaɯ]	[ʧanɯ]	[bebei]
		‘his/her foot’	‘his/her bell’	‘baby (ACC)’

2.2 Javanese

A parallel pattern in Javanese was first reported and analysed by Lee (1999; 2007). Table 2 shows the relevant alternations.[7] The definite suffix has an initial [n] after vowel-final stems ([kopine] ‘the coffee’) but not after consonant-final stems ([kuliṯe] ‘the skin’). A stem-final /h/ is deleted before a vowel-initial suffix ([səkolaan] ‘school building’ vs. [səkolah] ‘school’).[8] In /omah+ne/, n-deletion creates the intervocalic environment for h-deletion, but the deletion of /h/ obscures why /n/ was deleted in the first place (4). Therefore, n-deletion self-destructively feeds h-deletion.

	Bare	DEF *-(n)e*	*-an*
a. kopi ‘coffee’	[kopi]	[kopine]	
b. kulit ‘skin’	[kuliṯ]	[kuliṯe]	
c. omah ‘house’	[omah]	[omae]	[omaan]
d. sekolah ‘school’	[səkolah]	[səkolae]	[səkolaan]

Table 2: Javanese paradigms

(4) SDF in Javanese (Lee 1999; 2007)

	UR	a. /omah+ne/	b. /kuliṯ+ne/	c. /səkolah+an/
n-deletion:	n → ∅ / C_	∅	∅	
h-deletion:	h → ∅ / V_V	∅		∅
	SF	[omae]	[kuliṯe]	[səkolaan]
		‘the house’	‘the skin’	‘school building’

2.3 Japanese

The Japanese case has been described as ‘opaque allomorph selection’ (Hall et al. 2018). The nonpast suffix has an initial [r] after vowel-final verb stems ([taberu] ‘eat’) but not after consonant-final ones ([tobu] ‘fly’). Poser (1986; 1988) analysed the removal of /r/ after a consonant as Continuant Deletion. In addition, /w/ undergoes w-deletion before non-low vowels as in [kae] ‘buy!’ and [ie] ‘say!’ (Vance 1987; Nevins 2011), so /w/ only surfaces before [a] in native words in Japanese (e.g. [kawaseru] ‘make buy’).[9] Table 3

[7] Unless otherwise stated, the Javanese data come from personal elicitation and from the SEAlang Library Javanese Resources (http://sealang.net/java/corpus.htm).

[8] The suffix *-an* has multiple uses. Horne (1974) shows that it can be a nominaliser but the SEAlang Library Javanese Resources suggest that it can also form adjectives e.g. *omahan* ‘tame’, *sekolahan* ‘educated’. I have chosen not to gloss it, and the phonological facts remain unaffected.

[9] Unlike the other suffixes in the table, imperatives do not exhibit suffix-initial alternations. But how *-e* and *-ro* are suffixed to a stem is presented for two reasons. First, [kae] and [ie] demonstrate the effects of w-deletion alone, which is essential for establishing a constraint ranking in Japanese (Section 4.6). Second, they also demonstrate a preference of Japanese to select vowel-initial suffixes for consonant-final stems and consonant-initial suffixes for vowel-final stems, even for two underlyingly different allomorphs.

gives the relevant paradigms.[10] The suffix-initial continuants [r], [s] and [j] appear only after vowel-final stems, and the initial vowel [a] of the negative only after consonant-final stems. As for imperatives, *-e* appears after consonant-final stems and *-ro* appears after vowel-final stems. In /kaw+ru/, Continuant Deletion creates the environment for w-deletion, but the deletion of /w/ removes the consonant that motivated Continuant Deletion (5). Therefore, Continuant Deletion self-destructively feeds w-deletion.

	NPST *-(r)u*	NEG *-(a)nai*	CAUS *-(s)ase-ru*	IMP *-e*/*-ro*	COND *-(r)eba*	VOL *-(y)oo*
a. tabe- 'eat'	[taberu]	[tabenai]	[tabesaseru]	[tabero]	[tabereba]	[tabejoo]
b. mi- 'see'	[miru]	[minai]	[misaseru]	[miro]	[mireba]	[mijoo]
c. tob- 'fly'	[tobu]	[tobanai]	[tobaseru]	[tobe]	[tobeba]	[toboo]
d. tor- 'take'	[toru]	[toranai]	[toraseru]	[tore]	[toreba]	[toroo]
e. shin- 'die'	[ʃinu]	[ʃinanai]	[ʃinaseru]	[ʃine]	[ʃineba]	[ʃinoo]
f. kaw- 'buy'	[kau]	[kawanai]	[kawaseru]	[kae]	[kaeba]	[kaoo]
g. iw- 'say'	[iu]	[iwanai]	[iwaseru]	[ie]	[ieba]	[ioo]

Table 3: Japanese paradigms

(5) SDF in Japanese (Poser 1988; Hall et al. 2018)

		a. /kaw+ru/	b. /tob+ru/	c. /iw+e+ru/
	UR			
Continuant Deletion:	[+cont] → ∅ / C_	∅	∅	
w-deletion:	w → ∅ / _[–low]	∅		∅
	SF	[kau]	[tobu]	[ieru]
		'buy (NPST)'	'fly (NPST)'	'can say (NPST)'

2.4 Why SDF is a problem for Parallel OT

The four cases share a superficial property that explains why they count as overapplication. In the three consonantal cases, a sequence /VC+CV/ at a stem–suffix juncture surfaces as [VV]: both consonants are absent, although the absence of either one would suffice to avoid the consonant sequence. In the vowel case, a stem-final /VC/ sequence followed by a C-initial suffix is mapped onto [VVC#], where the epenthetic vowel that breaks the consonant cluster remains, but the consonant that would have formed part of the cluster disappears.

With standard segmental representations, Parallel OT cannot select these outputs. Take /ajak+sɯ/ for example (6). Elision can be attributed to a markedness constraint against consonant clusters (*CC) dominating MAX. Velar Deletion is the result of *VkV dominating MAX. The fact that the suffix instead of the stem consonant in a cluster is deleted can be captured by a root-specific faithfulness constraint, MAX_{root} (adapted from Wilson 2001: 174).[11] The attested [ajaɯ] incurs one MAX_{root} violation and two MAX violations. [ajasɯ] also incurs one MAX_{root} violation but only one MAX violation instead of two, and it violates no markedness constraint. The attested SF is therefore harmonically bounded.

[10] Unless otherwise stated, the Japanese words come from JMdict, the Japanese–Multilingual Dictionary of the Electronic Dictionary Research and Development Group (https://www.edrdg.org/jmdict/j_jmdict.html), and all Japanese forms have been confirmed by native speakers.

[11] In order to avoid confusion with the capital letters K and G used for underspecified velars later on, I write the constraint as *VkV to target both [k] and [g]. The voicing of the intervocalic velar does not impact the analysis.

(6) Standard segmental representations: the attested SF is harmonically bounded

/ajak+sɯ/	*CC	*VkV	MAX$_{root}$	MAX
a. ☹ ajaɯ			*	**!
b. 💣 ajasɯ			*	*
c. ajakɯ		*!		*
d. ajaksɯ	*!			

The point does not depend on the particular constraints. Any faithfulness constraint that penalises deletion, whether general, positional, or morpheme-specific, penalises the two deletions of [ajaɯ] at least as much as the single deletion of [ajasɯ]. A markedness constraint that favours [ajaɯ] would have to penalise post-vocalic [s], which is tolerated by other attested SFs such as [ʃiʃesi]. The same reasoning applies to Javanese [omae] and [omane] and to Japanese [kau] and [karu]. In the vowel case, [bebein] incurs a DEP and a MAX violation, whereas [beben] incurs only the MAX violation and [bebekin] incurs only the DEP violation and a *VkV violation.

3 Two properties shared by the SDF cases

The rules involved in the four interactions fall into two groups. The 'earlier' rule in each interaction (Epenthesis, Elision, n-deletion, Continuant Deletion) inserts or deletes a segment where a stem and a suffix meet to avoid a consonant cluster at the juncture.

(7) The earlier rule in each interaction

a. Epenthesis: ∅ → i / C_C#
b. Elision: s/j → ∅ / C_
c. n-deletion: n → ∅ / C_
d. Continuant Deletion: [+cont] → ∅ / C_

The 'later' rule (Velar Deletion, h-deletion, w-deletion) removes a stem-final segment that has become prevocalic or intervocalic through the application of the earlier rule.[12]

(8) The later rule in each interaction

a. Velar Deletion: k/g → ∅ / V_V
b. h-deletion: h → ∅ / V_V
c. w-deletion: w → ∅ / _[–low]

This section identifies two properties of these rules that previous discussions of SDF have not made explicit. First, each interaction involves a rule that only applies at morpheme junctures, and even there only to particular morphemes (Section 3.1). Second, they resolve stem–suffix consonant sequences in the same way, by realising a suffix-initial segment in required contexts (Section 3.2).

[12] I refer to the unit to which suffixes attach as the stem. In all the examples in this paper (except /iw+e+ru/) the stem is a root, but nothing in the analysis depends on this.

3.1 Alternation is restricted to morpheme junctures

The first property concerns the place at which these rules apply. A process shows nonderived environment blocking (NDEB) if it only applies in morphologically or phonologically derived environments (Kiparsky 1982; 1993). In the standard example, Finnish Assibilation (t → s / _i) only applies when the environment /ti/ is formed across morpheme concatenation boundaries (9a), or after its structural description is met by another rule (Vowel Raising; 9b), but not to a morpheme-internal /ti/ sequence (9d).[13]

(9) Finnish Assibilation

	UR	a. /halut+i/	b. /vete/	c. /sata/	d. /tilat/
Vowel Raising:	e → i / _#		i		
Assibilation:	t → s / _i	s	s		
	SF	[halusi]	[vesi]	[sata]	[tilat]
		‘wanted’	‘water’	‘hundred’	‘order’

Four of the deletion rules in the SDF cases demonstrably have this property: Turkish Velar Deletion and Elision and Javanese n-deletion and h-deletion do not apply inside a morpheme. Japanese Continuant Deletion also applies at morpheme junctures only, but it only qualifies as a case of NDEB if the morpheme-internal [Cj] sequences are considered clusters rather than palatalised segments (see below). Moreover, even at morpheme junctures, some of these rules apply only to particular morphemes.

Turkish Velar Deletion applies to stem-final velars that become intervocalic after suffixation (Table 1), but not to intervocalic velars inside a morpheme (Table 4). It does not even apply to all velars at morpheme junctures. Suffix-initial velars after vowel-final stems are never deleted (Table 5), and a number of polysyllabic nouns have stem-final velars that never alternate (Table 6). Inkelas (2011) shows that the alternation is further restricted to nominal and polysyllabic stems without a long final vowel. Lexical exceptions also vary between speakers. In other words, the environments in which Velar Deletion applies are morphologically derived, but it does not apply in *all* morphologically derived environments. Velar Deletion is thus both blocked in nonderived environments and lexically restricted in derived ones.

	SF	Gloss
a. avukat	[avukat]	‘lawyer’
b. hareket	[hareket]	‘movement’
c. vagon	[vagon]	‘railway car’
d. sigorta	[sigorta]	‘insurance’

Table 4: Turkish morpheme-internal intervocalic velar stops are preserved

Elision is restricted in the same way. Post-consonantal /s/ and /j/ are frequently found inside Turkish morphemes (Table 7). At junctures, Elision affects the initial consonant of the third person possessive *-(s)I* and the [j] of suffixes such as the first person predicative *-(y)Im*, but not the initial [s] of other suffixes (Table 8). For example, compare [ajaɯ]

[13] The Finnish example is used here only to illustrate the notion of NDEB as traditionally described. Anttila (2006) argues that Assibilation targets extrametrical coronal stops and can be stated without reference to derived environments (see also Anttila 2003). He also shows that Assibilation is restricted to certain suffixes and has lexical exceptions: not all /i/-initial suffixes trigger it (e.g. the conditional *-isi*), and some verbs never undergo it (Anttila 2006: 901, 920).

	SF	Gloss
a. seneki	[sene+ki]	‘this year’s’
b. adadaki	[adada+ki]	‘the one on the island’
c. altıgen	[altɯ+gen]	‘hexagon’
d. yedigen	[jedi+gen]	‘heptagon’

Table 5: Turkish suffix-initial velar stops are preserved after vowel-final stems

	Bare	ACC *-(y)I*	Gloss
a. orak	[orak]	[orakɯ]	‘sickle’
b. lâik	[laik]	[laiki]	‘secular’
c. antarktik	[antarktik]	[antarktiki]	‘Antarctic’
d. patolog	[patolog]	[patologu]	‘pathologist’
e. sinagog	[sinagog]	[sinagogu]	‘synagogue’

Table 6: Lexical exceptions to Turkish Velar Deletion (Inkelas 2011: 43)

‘his/her foot’ and [ajaksɯ] ‘foot-like’: the same stem is followed by an s-initial suffix with a high vowel, but only in [ajaɯ] does the suffix-initial [s] alternate. A parallel exists in predicative suffixes too: Elision applies to first person singular *-(y)Im* and plural *-(y)Iz* predicates ([masumum] ‘I am innocent’; [mutlujum] ‘I am happy’), but not to second person singulars ([gentʃsin] ‘you are young’; [yzgynsyn] ‘you are sad’).

	SF	Gloss		SF	Gloss
a. iksir	[iksir]	‘potion’	e. sübyan	[sybjan]	‘minor’
b. borsa	[borsa]	‘exchange’	f. madalya	[madaɬja]	‘medal’
c. afsun	[afsun]	‘spell’	g. isyan	[isjan]	‘rebellion’
d. konsey	[konsej]	‘council’	h. dünya	[dynja]	‘world’

Table 7: Turkish morpheme-internal post-consonantal /s/ and /j/

	Suffix	SF	Gloss
a. ayak-sı	*-sI* ‘-like’	[ajaksɯ]	‘foot-like’
b. durak-sız	*-sIz* ‘without’	[duraksɯz]	‘without stops’
c. gelenek-sel	*-sAl* ‘-al’	[geleneksel]	‘traditional’
d. genç-sin	*-sIn* ‘2SG.PRED’	[gentʃsin]	‘you are young’

Table 8: Turkish s-initial suffixes that do not undergo Elision after consonant-final stems

Likewise, n-deletion and h-deletion do not apply in Javanese morpheme-internally. Javanese has morpheme-internal sequences of a consonant followed by [n] (Table 9) and morpheme-internal intervocalic [h] (Table 10).[14]

In Japanese, Continuant Deletion of /r, s, j/ applies at morpheme junctures, but there is no strong evidence suggesting that it does not apply in nonderived environments, because the environment for Continuant Deletion cannot be found inside native or Sino-Japanese morphemes for [r] and [s]. In native morphemes, a coda can only be the first half of a geminate or a moraic nasal, and cannot be followed by [s] or a liquid (Itô 1986; Itô &

[14] Lee (1999; 2007) names the rule ‘n-deletion’ but uses the constraint *CN, which would cover all nasals. I keep the name ‘n-deletion’, since only coronal nasals are involved in SDF.

	SF	Gloss
a. sakwèhning	[sakʷɛhniŋ]	‘all’
b. prayitna	[prajɪt̪nɒ]	‘cautious, carefully’
c. ningnang	[niŋnaŋ]	‘no different, exactly the same’
d. pêrnah	[pərnah]	‘family; relationship’

Table 9: Javanese morpheme-internal consonant–nasal sequences

	SF	Gloss
a. dhihin	[dihin]	‘the first’
b. prihatin	[prɪhat̪ɪn]	‘anxious’
c. bihal	[bihal]	‘mule (smuggler)’ (borrowed from Arabic)
d. mihun	[mihun]	‘rice flour noodle’ (borrowed from Chinese)

Table 10: Javanese morpheme-internal intervocalic [h]

Mester 1995; Kurisu 2014). So, morpheme-internal [nr] or [ns] sequences do not arise. Consonant + [j] sequences exist and /j/ is preserved after a consonant inside morphemes (e.g. [bjoo] ‘second (of time)’). But this only supports an NDEB analysis of Continuant Deletion if such onsets are analysed as consonant clusters (Kubozono 2015; Nogita 2016; Hirayama & Vance 2018) rather than single palatalised segments (Itô & Mester 2003).

Furthermore, the effects of Continuant Deletion are observed only when certain suffixes are attached. Table 11 shows bimorphemic Sino-Japanese nouns with consonant + [r] sequences at morpheme boundaries. The [nr] sequences in these nouns are not reduced but those formed by the affixation of verbal suffixes are (cf. /ʃin+ru/ → [ʃinu] in Table 3). So, Continuant Deletion is restricted to particular verbal suffixes, rather than general at morpheme junctures.

	SF	Gloss
a. ken-ri	[kenri]	‘right (n.)’
b. nen-rei	[nenrei]	‘age’
c. kan-ren	[kanren]	‘connection’
d. shin-rai	[ʃinrai]	‘trust’

Table 11: Japanese consonant+/r/ sequences in Sino-Japanese words

The only process in SDF interactions that is language-general is Japanese w-deletion: [w] + non-low vowels are not found in Japanese except in some recent loanwords (Hall et al. 2018: 604). Epenthesis that breaks illicit coda clusters in Turkish generally exists (Sprouse 1997), but the vowel∼∅ alternation in SDF is different from it. Section 4.2 discusses these two different vowel alternations in more detail.

In short, the data in this section show that Velar Deletion, Elision, n-deletion, h-deletion and Continuant Deletion are only observed at morpheme junctures. Four of them fail to apply inside a morpheme, and so does Continuant Deletion if the palatalised onsets are consonant–glide clusters. Furthermore, even at junctures some of them apply only to particular morphemes: to some stem-final velars but not others, or to the suffix-initial continuants of only some suffixes. Which segments alternate is thus partly a lexical matter.

3.2 Suffix-initial segments are left unrealised

The second property concerns how the consonant sequences at stem–suffix junctures are resolved. In the three consonantal cases, the first rule (Elision, n-deletion, Continuant Deletion) applies when a suffix beginning with the target consonant (/s/, /j/, /n/ or /r/) follows a consonant-final stem. Descriptive work treats each suffix as having a consonant-initial variant after vowel-final stems and a consonant-less variant after consonant-final stems: [-sɯ]/[-ɯ] in Turkish, [-ne]/[-e] in Javanese and [-ru]/[-u] in Japanese (for Turkish, see van der Hulst & van de Weijer 1991; Hankamer 2011; for Javanese, Dudas 1976; Lee 1999; for Japanese, Kurisu 2012). In each case it is the second consonant of the stem–suffix sequence (C_2) (i.e. the one in the suffix) that is deleted (Tables 12–14).

	Gloss	Stem UR	Suffixation	SF
a. alay	'regiment'	/alaj/	alaj+sɪ	[alajɯ]
b. karar	'decision'	/karar/	karar+sɪ	[kararɯ]
c. çan	'bell'	/ʧan/	ʧan+sɪ	[ʧanɯ]
d. gülüş	'smile'	/gylyʃ/	gylyʃ+sɪ	[gylyʃy]
e. top	'ball'	/top/	top+sɪ	[topu]

Table 12: Turkish: C_2 is removed after suffixation of *-(s)ɪ*

	Gloss	Stem UR	Suffixation	SF
a. bénér	'right'	/bener/	bener+ne	[benere]
b. béring	'unbalanced'	/beriŋ/	beriŋ+ne	[beriŋe]
c. tégés	'significance'	/teges/	teges+ne	[tegese]
d. kuliʈ	'skin'	/kuliʈ/	kuliʈ+ne	[kuliʈe]

Table 13: Javanese: C_2 is removed after suffixation of *-(n)e*

	Gloss	Stem UR	Suffixation	SF
a. shir	'know'	/ʃir/	ʃir+ru	[ʃiru]
b. shin	'die'	/ʃin/	ʃin+ru	[ʃinu]
c. hanas	'speak'	/hanas/	hanas+ru	[hanasu]
d. tob	'fly'	/tob/	tob+ru	[tobu]

Table 14: Japanese: C_2 is removed after suffixation of *-(r)u*

Such alternations coincide with what Wilson (2001: 149) identifies as the one "very restricted" exception to the generalisation that intervocalic C_1C_2 clusters are simplified by deleting the first consonant: "some languages resolve illegal biconsonantal clusters that arise at root-suffix boundaries by deletion of the second (i.e. suffixal) consonant". Wilson attributed the pattern to a root-specific faithfulness constraint (Beckman 1998; Zoll 1998; McCarthy & Prince 1995) ranked above general MAX. In other words, suffix-initial segments are absent because of their suffixal status.

However, root-faithfulness is insufficient for SDF, because in the attested SDF output forms, the stem loses a segment as well ([ajaɯ], [omae], [kau]). As Section 4 shows, both the stem-final and the suffix-initial segments that alternate need to be lexically marked.

As for Turkish vowel-SDF, if the vowel is inserted, as on the rule analysis in (2), there is no reason to assign it to the suffix rather than to the stem: as Hockett (1947: 323) notes for an inserted vowel in Menomini, "Either cut will do; either choice is arbitrary." In the analysis developed below, the vowel belongs to the suffix because it is part of the

suffix's underlying representation. Why the alternating vowel is underlyingly in the suffix is discussed in detail in Section 4.2.

3.3 Summary

Both generalisations target segments at the edges of particular morphemes. The segments that alternate are lexically designated (some stem-final velars, some suffix-initial consonants), and alternation in all the processes except w-deletion is confined to morpheme junctures. Whether such a segment is realised depends on the adjacent morpheme: a suffix-initial consonant appears after a vowel, a suffix-initial vowel appears after a consonant, and a stem-final velar before a consonant or at the end of the word. The analysis in Section 4 encodes the lexical specification of segments in the underlying representation and derives the context-dependent realisation from constraint interaction. The shared properties mean that the SDF patterns follow from the same grammar, with no further assumptions in the consonantal cases.

4 Analysis

4.1 Underspecified segments

As shown in the previous section, whether a segment alternates is an unpredictable property of individual morphemes. In Turkish, there are stem-final velars that alternate ([bebek] ∼ [bebei]) as well as those that do not ([orak] ∼ [orakɯ]; Table 6). The same can be found for s-initial suffixes: *-(s)I* alternates depending on the stem-final segment, but the similative always surfaces with [s] at the beginning (Table 8). In Japanese, verbal suffixes have alternating /r/ ([taberu] ∼ [tobu] for *-(r)u*), but Sino-Japanese morphemes have non-alternating /r/ ([kenri]; Table 11). I argue that this is encoded in the lexicon by the contrast between underspecified and fully-specified segments.

I assume that URs are selected by Lexicon Optimisation (Prince & Smolensky 1993) in the alternation-sensitive way proposed by Inkelas (1995). As per Richness of the Base (Prince & Smolensky 1993), a segment at any position in any morpheme can be either under- or fully-specified, and the grammar must map every such input to an output. Lexicon Optimisation determines which of these inputs the learner chooses to be the UR. Where both inputs give the same output, Lexicon Optimisation chooses the UR that leads to the most harmonic mapping. So, a non-alternating segment is always represented as fully-specified even where its presence is predictable from the context, in contrast with classical underspecification where predictable information is left out (see also Inkelas et al. 1997).[15]

The current analysis builds on the underspecification approach to NDEB proposed by Kiparsky (1993) and further developed by Rasin (2025). In his autosegmental underspecification account of Finnish consonant gradation, Kiparsky (1993) lets the presence of part of a geminate depend on its association to a C slot on the skeletal tier. I adopt this idea, but not his representation of geminates, and apply it to whole segments, as stated in (10).

[15] Gorman & Reiss (2026) also analysed segment deletion rules like Turkish Velar Deletion with underspecification, but in a counterintuitive way. They make the deleting velars more richly specified than the non-deleting ones. Although their proposal differs from the present one in the choice of underspecified segments, both analyses show that alternation patterns need to be explained with the contrast between segments encoded in the lexicon.

(10) Autosegmental underspecification account

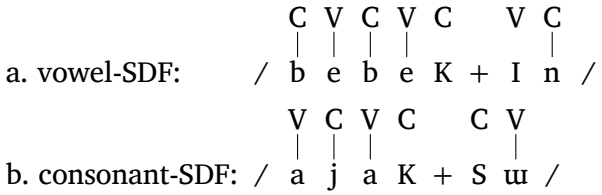
a. An alternating segment is *underspecified*: it is present in the input as a melodic unit, with its features (including whether it is consonantal or vocalic), but it is not associated to a C or V slot on the skeletal tier.

b. A non-alternating segment is fully-specified and associated to a C or V slot in the input.

c. A segment is pronounced if and only if it is associated in the output.

Underspecified segments of this kind are familiar from analyses of 'ghost' or 'latent' segments, which are present underlyingly but surface only in some contexts (e.g. Zoll 1996; Zimmermann 2019).[16] I write them with capital letters. The stem-final velar of /bebeK/ 'baby' is underspecified, and so are the initial vowel of the possessive suffix /-In/ and the initial consonant of /-SI/.[17] The inputs for vowel-SDF and consonant-SDF are shown in (11) as examples, and Table 15 lists the inputs for all four cases.

(11) Underspecified segments in the Turkish SDF inputs



	Input	Output
a. Turkish vowel-SDF	/bebeK+In/	[bebein]
b. Turkish consonant-SDF	/ajaK+Sɯ/	[ajaɯ]
c. Javanese	/omaH+Ne/	[omae]
d. Japanese	/kaW+Ru/	[kau]

Table 15: Underspecified segments in the inputs of the four SDF cases

In an output candidate, an underspecified segment can be linked to a slot on the skeletal tier, in which case it is pronounced. It can also remain unlinked, in which case it is silent, or it can be deleted. A segment unassociated with a C/V slot sounds the same as ∅, but they incur different violations (Section 4.3). A fully-specified input segment can likewise stay associated, lose its association or be deleted. For example, [bebeKin] is pronounced [bebein], but it retains the stem-final melody.

Finally, although the underspecified segments listed in Table 15 are all found at morpheme junctures, it should be noted that nothing in the lexicon states that alternating segments occur only at morpheme edges. Where an alternation can occur is derived by the grammar, and as I show in Section 4.7, underspecified segments elsewhere could

[16] Other ways of encoding the lexical contrast in the representation of inputs exist too. An anonymous reviewer suggests Gradient Symbolic Representations (GSR; Smolensky & Goldrick 2016). In GSR, positions can hold symbols with a partial degree of presence, determined by their 'activity'. Ordinary segments have activity 1 and 'ghost' segments have activity between 0 and 1. A weakly active ghost (activity < 0.5) surfaces only when it helps avoid a markedness violation, and a strongly active one is lost only to avoid one (Zimmermann 2019). This approach could plausibly be applied to the 'ghost' segments here. The binary contrast between underspecified and fully-specified segments used in the present analysis might be preferable because it suffices for the SDF cases, and it does not require arbitrary numerical activation values or weighted constraints. Every distinction it makes is one of standard autosegmental representation.

[17] The capital I in Turkish suffixes is often used for a high vowel unspecified for the features determined by vowel harmony. The vowel in /-SI/ is also underspecified, but not in the autosegmental sense in (10).

never alternate. Section 6.1 compares this treatment of the lexicon with Rasin's (2025) morpheme structure constraints.

4.2 The Turkish alternating vowel

Under the analysis proposed here, the vowel in the possessive /-In/ is present in the input rather than inserted. This echoes the classical analysis of Lees (1961: 40), who posits an underlying suffix-initial vowel that is deleted after vowels. Hankamer (2011) argues instead that the vowel is epenthetic. His observation is that the suffixes with an initial vowel∼∅ alternation (the possessives *-(I)m* and *-(I)n*, the reflexive *-(I)n* and the reciprocal *-(I)ş*) are single consonants, and that the vowel appears exactly where the suffix consonant would otherwise form an illicit cluster with a consonant-final stem. Hankamer (2011) also argues that the epenthesis proposal explains why the progressive *-Iyor*, the only productive vowel-initial suffix, rather than the stem, keeps its vowel in a VV sequence at junctures. A suffix such as *-(I)mIz* 'our', whose second vowel is not needed to avoid a cluster, is analysed by Hankamer (2011: §4.4) as two suffixes with cyclic epenthesis.

Whether the vowel is epenthetic affects whether vowel-SDF can be accounted for in the present analysis. If the vowel is epenthetic, no faithfulness constraint can protect a vowel that is absent from an input like /bebeK+n/. The analysis below then covers the three consonantal cases, but not vowel-SDF. If the vowel is underlying, vowel-SDF has the same status as consonant-SDF by having suffix-initial segments that alternate depending on the preceding segment. I adopt the underlying-vowel analysis for the present analysis. This means that Hankamer's generalisation describes a coincidence of the lexicon. The suffixes with an underspecified initial vowel happen to contain a single consonant (or two, in *-(I)mIz* and *-(I)nIz*) besides the vowel. Since my central claims do not depend on vowel-SDF, readers who prefer the epenthesis analysis can take the vowel case as a conditional extension of the account.

Finally, it should be noted that vowel∼∅ alternations inside roots, as in [akɯl] 'mind' ∼ [aklɯ] 'his/her mind', come from a different process. As Section 4.7 shows, a stem-internal underspecified vowel could not alternate under the present grammar. These alternations are analysed as epenthesis triggered by an illicit final cluster (Lees 1961: 37–38; Hankamer 2011: §4.5), which is unaffected by the assumptions adopted in the present proposal.

4.3 Constraints

The complete constraint set is given in (12). The same constraints are used for all three languages, except that each language has its own markedness constraint against the stem-final segment in the relevant context: *VkV in Turkish, *VhV in Javanese and *w[–low] in Japanese.

(12) Constraints

a. SPECIFY: Assign one violation for each melodic unit in the output that is not associated to a C or V slot (cf. SPECIFY[T]; Myers 1997: 861; Zoll 2003: 241).

b. MAXLINK: Assign one violation for each association line in the input that has no correspondent in the output (cf. PARSELINK; Itô et al. 1995: 586).

c. DEPLINK: Assign one violation for each association line in the output that has no correspondent in the input (cf. FILLLINK; Itô et al. 1995: 586; NOLINK; McCarthy 2008: 278).
d. $\text{DEPLINK}_{\text{suffix}}$: Assign one violation for each association line in the output that links a melodic unit of a suffix and has no correspondent in the input.
e. MAX: Assign one violation for each melodic unit in the input that has no correspondent in the output.
f. MAX-C/V_V: Assign one violation for each consonantal unit that is between two vocalic units in the input and has no correspondent in the output.
g. MAX-V/C_C: Assign one violation for each vocalic unit that is between two consonantal units in the input and has no correspondent in the output.
h. *VkV (Turkish): Assign one violation for each pronounced [k] or [g] between two pronounced vowels.
*VhV (Javanese): Assign one violation for each pronounced [h] between two pronounced vowels.
*w[–low] (Japanese): Assign one violation for each pronounced [w] followed by a pronounced non-low vowel.

Besides the general markedness constraints in (12h), the other constraints fall into two groups. The first group consists of SPECIFY, MAXLINK, DEPLINK, $\text{DEPLINK}_{\text{suffix}}$, and MAX, all of which evaluate association. An underspecified segment that is realised violates DEPLINK. One that is deleted violates MAX, and one that is retained but left unassociated violates SPECIFY. A fully-specified segment that is deleted or delinked violates MAXLINK. MAXLINK thus distinguishes non-alternating from alternating segments: deleting a fully-specified segment destroys an association line, but deleting an underspecified segment does not. A constraint against deleting fully-specified segments (MAX_{full}) would do the same work for deletion, but MAXLINK penalises delinking in addition to deletion, and it pairs naturally with DEPLINK. With MAXLINK, intervocalic velars of /avukat/ and /bebeK+i/ that differ only in association can be both derived in a grammar (see Section 4.4 and Appendix B).

$\text{DEPLINK}_{\text{suffix}}$ is a morpheme-specific version of DEPLINK (McCarthy & Prince 1995; Beckman 1998). Every violation of $\text{DEPLINK}_{\text{suffix}}$ is also a violation of DEPLINK, so associating a suffix segment is always at least as costly as associating a stem segment. As shown in Section 6.2, this stringency relation excludes grammars in which underspecified stem segments are lost in a context where underspecified suffix segments survive. It thus derives the asymmetry of Section 3.2 without a root-specific constraint that would penalise the stem deletions in SDF.[18]

The second group consists of the two contextual faithfulness constraints, MAX-C/V_V and MAX-V/C_C, whose contexts are defined with respect to the input. These two constraints preserve vowels between two consonants and consonants between vowels in the input, with morpheme boundaries ignored. The two neighbouring units can be either under- or fully-specified. For example, in /bebeK+In/, removal of /I/ violates MAX-V/C_C because /I/ occurs between two consonants in the input (/K/ and /n/). By contrast, the markedness constraints evaluate pronounced output segments. Contextual faithfulness finds its root in positional faithfulness (Beckman 1998). Such faithfulness constraints with input-defined contexts have been used by Steriade (2008: 171) in her P-map theory

[18] $\text{DEPLINK}_{\text{suffix}}$ outranks general MAX and DEPLINK (Section 4.4). This does not contradict the meta-ranking of root faithfulness above affix faithfulness (McCarthy & Prince 1995), insofar as it also conveys the idea that suffix segments are lost more easily than stem segments. For cases where affix faithfulness takes precedence over root faithfulness, see Rolle (2018) and Noske (2000); cf. Walker (2005).

to decide which consonant is removed in a cluster. In a V-stop-strident-stop-V cluster, stridents are preserved at the cost of postvocalic stops because the postvocalic stop∼∅ contrast is less perceptible. Steriade (2008: 171) captured the pattern with CV// ≫ MAXstrident/C_C ≫ MAX[−cont]/V_C ≫ MAX[−cont]/C_C.[19] Constraints of this kind have been used in analyses of opacity by Hauser & Hughto (2020), who derived several kinds of underapplication in Parallel OT and overapplication in Harmonic Serialism. The present constraints are less stipulative in the sense that they capture the preference for alternating CV patterns in Turkish, Javanese and Japanese (van der Hulst & van de Weijer 1991; Yip 1989; Vance 2008), which is also favoured crosslinguistically.

The work of contextual faithfulness is often done by phonotactic constraints on syllable structure, but this is not possible for SDF. [ajaɯ], [bebein], [omae] and [kau] all contain a VV sequence that output-oriented constraints such as ONSET would penalise. [bebein] also lacks the consonant whose presence would justify keeping the vowel. Only a constraint that refers to the input context can preserve /I/ in /bebeK+In/ after /K/ has been lost, or fail to protect /S/ in /ajaK+Sɯ/ although its output environment is intervocalic. Appendix B confirms this computationally. Section 6.2 returns to the worry that contextual faithfulness is too expressive.

The analysis does not cover every possible candidate: insertion of new melodies, changes of features, reordering and multiple association are assumed to be excluded by undominated constraints that are not shown. Each input melodic unit is associated, unassociated or deleted in a candidate. The tableaux below only show those most critical to SDF; all rankings were verified computationally over the full candidate space defined here, as described in Appendix B.

4.4 Turkish: ranking arguments

This section establishes the Turkish ranking from the bottom up, considering the simple alternations first and the SDF forms last. All tableaux contain the complete constraint set in a total order compatible with all of the ranking arguments. Figure 1 at the end of Section 4.5 summarises the resulting partial order.

The bare stem /bebeK/ surfaces with its final velar (13). Deleting /K/ violates MAX, and leaving it unassociated violates SPECIFY, whereas realising it violates only DEPLINK. Hence MAX ≫ DEPLINK. The same ranking preserves /K/ before a suffix with a fully-specified initial consonant, such as the plural /-ler/.

(13) Underspecified stem-final consonants are realised word-finally and before consonants: MAX ≫ DEPLINK

a.

/bebeK/	SPECIFY	MAX-V/C_C	MAXLINK	*VkV	MAX-C/V_V	$\text{DEPLINK}_{\text{suffix}}$	MAX	DEPLINK
a. ☞ bebek								∗
b. bebe							∗!	
c. bebeK	∗!							

b.

[19] McCarthy (2009) points out that the phonetic information to which these constraints refer may not be always present in the input. Since the constraints in (12) refer only to the C/V status of input neighbours, which is in the input, this objection is not applicable here.

/bebeK+ler/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ bebekler								*
b. bebeler							*!	
c. bebeKler	*!							

An underspecified suffix-initial consonant is not realised after a consonant-final stem (14). Realising /S/ in /ʧan+Sɯ/ violates both DepLink$_{suffix}$ and DepLink, whereas deleting it violates Max. Since Max dominates DepLink, DepLink$_{suffix}$ must dominate Max. Deleting the stem-final /n/ instead is fatal because /n/ is fully-specified. The same ranking deletes the underspecified vowel of /-In/ after a vowel-final stem.

(14) Underspecified suffix-initial segments are not realised by default: DepLink$_{suffix}$ ≫ Max

a.

/ʧan+Sɯ/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ ʧanɯ							*	
b. ʧansɯ						*!		*
c. ʧasɯ			*!			*	*	*
d. ʧaɯ			*!				**	
e. ʧanSɯ	*!							

b.

/anne+In/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ annen							*	
b. annein						*!		*
c. annin			*!			*	*	*
d. anneIn	*!							

After a vowel-final stem, the underspecified consonant is realised (15). In /ʃiʃe+Si/, /S/ is between two vowels in the input, so deleting it violates Max-C/V_V. This requires Max-C/V_V ≫ DepLink$_{suffix}$. Similarly, the underspecified vowel of /ip+In/ is realised between two consonants, which requires Max-V/C_C ≫ DepLink$_{suffix}$.

(15) Underspecified segments in a CV-alternating input context are realised: Max-C/V_V, Max-V/C_C ≫ DepLink$_{suffix}$

a.

/ʃiʃe+Si/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ ʃiʃesi						*		*
b. ʃiʃei					*!		*	
c. ʃiʃeSi	*!							

b.

/ip+In/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ ipin						*		*
b. ipn		*!					*	
c. ipIn	*!							

An underspecified stem-final velar is not realised before a vowel-initial suffix (16). In /bebeK+i/, /K/ is flanked by vowels in the input, so its deletion violates Max-C/V_V as well as Max. But realising it violates *VkV, which must therefore dominate Max-C/V_V. The candidate [bebeKi], where /K/ remains underspecified, shows that Specify must also dominate Max-C/V_V.

(16) Underspecified stem-final velars are not realised between vowels: *VkV, Specify ≫ Max-C/V_V

/bebeK+i/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ bebei					*		*	
b. bebeki				*!				*
c. bebeKi	*!							

Fully-specified velars, by contrast, survive in the same environment (17). The velar of /avukat/ is intervocalic, just as /K/ in /bebeK+i/ is, so *VkV penalises both. However, only deleting or delinking /k/ in /avukat/ violates MaxLink, which must therefore dominate *VkV. The same ranking protects the suffix-initial velar of /sene+ki/ (Table 5) and the non-alternating stem-final velar of /orak+ɯ/ (Table 6). NDEB is thus captured in this way: MaxLink preserves the fully-specified segments.

(17) Fully-specified velars are preserved between vowels: MaxLink ≫ *VkV

a.

/avukat/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ avukat				*				
b. avuat			*!		*		*	
c. avuKat	*!		*					

b.

/sene+ki/	Specify	Max-V/C_C	MaxLink	*VkV	Max-C/V_V	DepLink$_{suffix}$	Max	DepLink
a. ☞ seneki				*				
b. senei			*!		*		*	
c. seneKi	*!		*					

These simple alternations establish every ranking of the Turkish grammar except one, Max-V/C_C ≫ Max-C/V_V, which only the vowel-SDF form requires (Section 4.5).

4.5 SDF without additional machinery

The SDF forms can now be evaluated with the ranking that the simple alternations have established. In /ajaK+Sɯ/, neither underspecified consonant is flanked by vowels in the

input (18), so the contextual faithfulness constraint protects neither of them. Realising /S/ violates DEPLINK$_{suffix}$ whether or not /K/ is realised ([ajasɯ], [ajaksɯ]), just as in [tʃansɯ]. Realising only /K/ yields an intervocalic velar ([ajakɯ]), just as in [bebeki]. Therefore, [ajaɯ] wins with two MAX violations. The second tableau in (18) shows that the same stem keeps its velar before the similative suffix *-sI* [ajaksɯ], whose initial consonant is fully-specified.

(18) Turkish consonant-SDF and its non-SDF counterpart derived

a.

/ajaK+Sɯ/	SPECIFY	MAX-V/C_C	MAXLINK	*VkV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ ajaɯ							**	
b. ajasɯ						*!	*	*
c. ajaksɯ						*!		**
d. ajakɯ				*!			*	*
e. ajaKSɯ	*!*							

b.

/ajaK+sɯ/	SPECIFY	MAX-V/C_C	MAXLINK	*VkV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ ajaksɯ								*
b. ajasɯ							*!	
c. ajaɯ			*!				**	

The comparison with the standard analysis in (6) shows why the current analysis works. Without underspecification, deleting the underlying suffix-initial /s/ is more costly than keeping it, so the transparent candidate [ajasɯ] harmonically bounds the SDF output [ajaɯ]. Now, with underspecification, realising the suffix-initial /S/ is the more costly option because it violates DEPLINK$_{suffix}$, whereas deleting it violates the more lowly ranked MAX. Realising the stem-final velar instead violates the even more highly ranked *VkV. So, given that neither consonant is in an intervocalic environment that protects it, the optimal choice is to delete both, which leads to the SDF form [ajaɯ].

For vowel-SDF, one further ranking is needed (19). In /bebeK+In/, /I/ is flanked by consonants and /K/ by vowels in the input. [bebein] deletes /K/ and realises /I/, so MAX-V/C_C needs to outrank MAX-C/V_V. The transparent candidate *[beben] loses because /I/ is protected by its input context, which does not change even when /K/ is deleted.

(19) Turkish vowel-SDF: MAX-V/C_C ≫ MAX-C/V_V

/bebeK+In/	SPECIFY	MAX-V/C_C	MAXLINK	*VkV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ bebein					*	*	*	*
b. beben		*!			*		**	
c. bebekn		*!					*	*
d. bebekin				*!		*		**
e. bebeKin	*!					*		*
f. bebeKIn	*!*							

In this way, SDF output is in fact a consequence of the simple alternations. Every total order of the constraints that derives the forms in Section 4.4 also selects [ajaɯ] for /ajaK+Sɯ/; see the results of computational verification in Appendix B. That is, any grammar that has Velar Deletion and Elision analysed in this way also naturally has consonant-SDF. The same holds for the consonantal SDF cases in Javanese and Japanese (Section 4.6), as well as for vowel-SDF in Turkish, with the one additional ranking established above.

Up to this point, all the rankings of the Turkish grammar have been established. Figure 1 summarises them in a Hasse diagram.

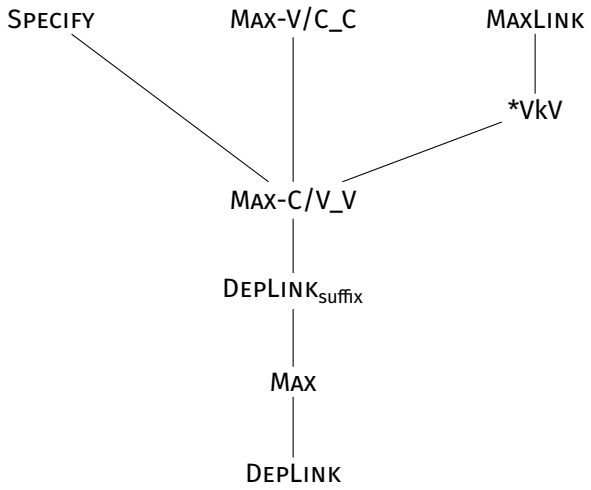


Figure 1: Hasse diagram of the Turkish ranking

4.6 Javanese and Japanese

The analysis carries over to Javanese and Japanese with the language-specific markedness constraints *VhV and *w[–low]. The tableaux are given in Appendix A, and the rankings are summarised in (20).

(20) Rankings established by the Javanese and Japanese data

a. Javanese: MaxLink ≫ *VhV ≫ Max-C/V_V; Specify ≫ Max-C/V_V ≫ DepLink$_{suffix}$ ≫ Max ≫ DepLink

b. Japanese: {*w[–low], Specify, MaxLink} ≫ Max-C/V_V ≫ DepLink$_{suffix}$ ≫ Max ≫ DepLink; Max-V/C_C ≫ DepLink$_{suffix}$

The Javanese ranking matches the Turkish one, except that Max-V/C_C is not shown because there is no vowel∼∅ alternation in the relevant suffixes to determine its position. The SDF output [omae] naturally falls out from the grammar generating the simple alternations.

Much of the Japanese ranking also matches the Turkish one, with three exceptions. First, the ranking between Max-V/C_C and Max-C/V_V is unresolved. The fact that the initial vowel of the negative suffix /-Anai/ is present after consonant-final stems ([tobanai] ‘not fly’) but absent after vowel-final ones ([tabenai] ‘not eat’) only establishes Max-V/C_C ≫ DepLink$_{suffix}$ (29), parallel to the Turkish suffix /-In/. But there is no context in which (i) an underspecified stem-final consonant and suffix-initial vowel meet

and (ii) the former is deleted while the latter is retained, as in /bebeK+In/ → [bebein]. Therefore, one cannot draw the conclusion that MAX-V/C_C ≫ MAX-C/V_V (19).

Second, although MAXLINK ≫ MAX-C/V_V in Japanese, this relationship is not derived by transitivity. In Turkish and Javanese, the markedness constraints outrank MAX-C/V_V because of the deletion of stem-final velars and /H/ (16), (26), and MAXLINK ≫ the markedness constraints because fully-specified velars and /h/ are preserved intervocalically (17), (26). By transitivity, MAXLINK ≫ MAX-C/V_V. In Japanese, this is established by the imperative [kae] 'buy!': when a vowel-initial suffix is attached, the stem-final /W/, instead of the fully-specified /e/, is deleted, although the deletion of either would satisfy *w[–low] (31).[20]

Third, the Japanese data do not rank MAXLINK with respect to *w[–low]. Unlike the consonant deletion processes in Turkish and Javanese, which only apply at morpheme junctures, w-deletion is general: there is no [w] + non-low vowel sequence found in native Japanese vocabulary. So MAXLINK does not need to dominate *w[–low] to ensure that fully-specified /w/ is not deleted. The recent loanwords in which [w] precedes non-low vowels can be handled by the stratum-specific faithfulness constraints that are independently needed for the Japanese loanword phonology (e.g. Itô & Mester 1999).

4.7 Where alternations can occur

An anonymous reviewer pointed out that the underspecification analysis of NDEB is at risk of circularity: a segment alternates because it is underspecified, and it is underspecified because it alternates. This is true if underspecification is the only determining factor for where an alternation occurs. The present account does not run into this problem, because the UR only encodes which segments can alternate, while the locations of alternation are derived by the grammar. That is, representing a segment as under- or fully-specified is necessary but not sufficient to decide its alternation or non-alternation.

Take Turkish for example. Morpheme-internal velars (/avukat/), morpheme-internal vowels (/afsun/), suffix-initial velars (/-ki/), stem-final non-velars (/tas/ 'bowl') and morpheme-internal post-consonantal /s/ (/borsa/) are represented as fully-specified because of the absence of alternation. Should they be underspecified, /borSa/ would always surface as [borsa] because MAX ≫ DEPLINK, /afsUn/ always as [afsun] because MAX-V/C_C ≫ DEPLINK, and /avuKat/ always as [avuat] because *VkV ≫ MAX-C/V_V. Predictions for these segments would not change because the context in which they appear does not vary across the paradigm. The contexts for /taS/ and /-Ki/ vary depending on the attaching morpheme. When followed by a consonant-initial suffix (/taS+lar/), /S/ is predicted to surface ([taslar]) because MAX ≫ DEPLINK; before vowel-initial suffixes (/taS+ɯ/), /S/ is also predicted to remain ([tasɯ]) because it does not violate the more highly ranked markedness constraint *VkV. In intervocalic environments (/sene+Ki/), /K/ is always removed because *VkV ≫ MAX-C/V_V; when it follows a stem-final consonant (/ip+Ki/), /K/ is still deleted because $\text{DEPLINK}_{\text{suffix}}$ ≫ MAX. So, no alternation would be found in any of these cases. The same applies to similar segments in Javanese and Japanese.

More importantly, such underspecified segments would be indistinguishable from fully-specified ones or no segment at all. Both /borSa/ and /borsa/ lead to [borsa], and both /taS/ and /tas/ lead to [tas], so speakers of Turkish would never posit /borSa/ because

[20] Technically, REALIZEMORPHEME (Kurisu 2001), which penalises a morpheme that has no exponent in the output, would also exclude *[kaw], in which the imperative suffix /-e/ is deleted, if it dominated MAX-C/V_V. If REALIZEMORPHEME were included in the constraint set, the Japanese data would therefore establish only that MAXLINK or REALIZEMORPHEME dominates MAX-C/V_V.

it incurs more DEPLINK violations. Similarly, the underspecified velar in /avuKat/ and /sene+Ki/ would be indistinguishable from no segment, which learners would not posit under Lexicon Optimisation either. The fact that both are pronounced means that they must be fully-specified, and their non-alternation is derived rather than prespecified. In this way, underspecification is only necessary for alternating segments, and alternation is only found at places where the context in which the segment appears differs across the paradigm, i.e. in derived environments. NDEB is accounted for without stipulation.

In brief, the account is circular only in the trivial sense that any analysis must record the distinction between alternating and non-alternating segments in positions where alternation is possible. As Inkelas (2011) showed, this distinction cannot be reduced to derived-environment conditions and it varies between speakers. The positional restrictions, the absence of suffix-initial velar deletion, and the SDF outputs all follow from the grammar.

5 Comparison with previous analyses

Section 2.4 showed why SDF is a problem for Parallel OT with standard representations. This section considers how other analyses deal with SDF and the two properties identified in Section 3. Serial and quasi-serial analyses capture SDF, but they need additional (sometimes unconventional) assumptions for NDEB or give up strict parallelism (Section 5.1.1). Section 5.1.2 discusses analyses that enrich output representations. Analyses of NDEB in Parallel OT capture the edge-only nature of some rules but leave SDF unexplained; they are discussed in Section 5.2.

5.1 Previous analyses of SDF

5.1.1 Serial and quasi-serial analyses of SDF

Rule-based serialism models SDF directly (2)–(5). When combined with underspecification (Kiparsky 1993; Rasin 2025), it also captures NDEB. As exemplified in (21), context-specific rules apply first to underspecified segments only, followed by the default rules to all the segments still underspecified at that point. Only underspecified segments in the specified context are affected by the feature-filling (or association-changing) rules. Those outside of the structural descriptions of context-specific rules are always realised by the default rules. Fully-specified segments are not affected.

(21) Rule-based serialism with underspecification

	UR	a. /ajaK+Sɯ/	b. /bebeK+ler/	c. /avukat/
Elision:	S/J → ∅ / C_	∅		
Velar Deletion:	K/G → ∅ / V_V	∅		
Default:	S/J → s/j			
Default:	K/G → k/g		k	
	SF	[ajaɯ]	[bebekler]	[avukat]
		‘his/her foot’	‘baby (PL)’	‘lawyer’

Rules with underspecified segments succeed on all three consonantal SDF cases, if the focus of each rule applying at certain morpheme junctures is rewritten as underspecified and the rules retain their relative order. Vowel-SDF can also be accounted for as long as the Epenthesis rule is replaced by a rule that realises the underlying /I/, which is ordered

before Velar Deletion. So, once it is clarified how segments are represented underlyingly, both serial and parallel approaches are able to capture SDF.

Sympathy (McCarthy 1999) and *OT-CC* (McCarthy 2006) have both been applied to Javanese SDF. Sympathy accommodates opacity in OT by "selecting a failed candidate to influence the output, and exercising that influence through a kind of faithfulness of the output to this failed candidate" (McCarthy 1999: 335). On Lee's (1999) Sympathy analysis, [omae] wins over [omane] because the sympathy constraint penalises segment insertion with respect to [omahe], the most harmonic candidate that does not violate the selector constraint $\textsc{Max}_{\text{root}}$. OT-CC evaluates chains of candidates that gradually diverge from the initial fully-faithful form with harmonic improvement (McCarthy 2006: 14), and captures opacity by restricting the order in which faithfulness violations appear in a chain using PREC constraints. Under Lee's (2007) analysis, the candidate chain in which $\textsc{Max}_{\text{root}}$ is violated without a prior violation of $\textsc{Max}_{\text{suffix}}$ is disfavoured, so [omane] loses to [omae]. If markedness constraints may refer to morpheme boundaries, both analyses can describe NDEB as well. However, both solutions were implemented in a way very similar to rule-based serialism. They had to refer to either a failed candidate, which is essentially the result of applying the earlier process in the rule interaction, or the steps of a candidate chain in a derivation; neither strictly adheres to the parallelism of OT. In addition, Rasin (2025) argues that candidate chains do not by themselves provide an adequate theory of NDEB.

Harmonic Serialism with contextual faithfulness offers another serial route. GEN is limited to making one change at a time and candidates are evaluated against the same constraint ranking in a series of steps (McCarthy 2010). Hauser (2016: §5.3) analyses Turkish consonant-SDF with a contextual faithfulness constraint MAX-g/_C to protect a velar that precedes a consonant in the input in each step. Consequently, /s/ is removed in the first step because /k/ precedes /s/ in /ajak+sɯ/. In the second step, the input is /ajak+ɯ/, and /k/ is removed because it violates *VkV and is no longer protected by MAX-g/_C. The present analysis fares better on two counts. First, it accounts for SDF in a single parallel input–output evaluation; no intermediate steps as in Harmonic Serialism are needed. Second, the contextual faithfulness constraints used in this paper are better justified. Turkish, Javanese and Japanese all favour the CV-alternating patterns on the surface, and preserving intervocalic consonants and interconsonantal vowels in languages that avoid clusters is crosslinguistically common. In contrast, keeping a velar because it appears before a consonant cannot be independently motivated in Turkish.

For the Turkish SDF cases, *level ordering* is a possible alternative. Inkelas & Orgun (1995) motivate at least four lexical levels in Turkish and propose that a form undergoes only the phonology of the levels at which it is morphologically derived. So, Elision and Velar Deletion might be restricted to particular suffixes at certain levels. Velar Deletion only applies when vowel-initial suffixes from levels 3 and 4 are attached (Inkelas & Orgun 1995). However, the level of the relativiser *-ki* is unclear, because it could appear between suffixes from levels 1 and 3, and between those from levels 3 and 4 (Inkelas & Orgun 1995: 769). In other words, level ordering cannot determine its behaviour. The analysis based on specification avoids this problem: *-ki* does not alternate because /k/ is lexically fully-specified and, as shown in Section 4.7, it could not alternate even if it were underspecified.

Elision raises a different problem. Level ordering predicts that suffixes from the same level, as long as they meet the structural description of a rule, should undergo a certain process in the same manner. However, there are suffixes in Turkish that belong to the same level but undergo the same process differently. The predicative second person sin-

gular is *-sIn* '2SG.PRED', which does *not* undergo Elision (e.g. [gentʃsin] 'you are young'; [yzgynsyn] 'you are sad'). In contrast, the first person singular and plural forms, *-(y)Im* '1SG.PRED' and *-(y)Iz* '1PL.PRED' respectively, *do* undergo Elision (e.g. [masumum] 'I am innocent' vs. [mutlujum] 'I am happy'). Although not explicitly mentioned in Inkelas & Orgun (1995), the predicative suffixes should all belong to level 4 as agreement suffixes (cf. Inkelas & Orgun 1995: 765). However, they undergo the same process differently, contrary to this prediction. This means that the alternation observed is a property of individual suffixes rather than levels.

5.1.2 Representational methods

Turbidity (Goldrick & Smolensky 1999; Goldrick 2000) distinguishes segments that are projected from those that are pronounced on the surface; a projected but not pronounced segment is silent, as if it were deleted. In Baković's (2007: 229–230) analysis of vowel-SDF, the winning candidate [bebe̲kin] contains a velar that is projected but not pronounced (underlined, following Baković 2007: 230). NOPROJCOMPLEX bans projected consonant clusters. NOPRONV[k]+V penalises pronounced morpheme-final velars that are made intervocalic. [bebe̲kin] violates neither: a consonant cluster is avoided by the epenthetic [i] and the intervocalic [k] is not pronounced. As an anonymous reviewer points out, reference to a morpheme boundary allows NOPRONV[k]+V to account for NDEB: it is not violated by the morpheme-internal velar of [avukat], nor by the suffix-initial velar of [seneki]. But forms such as [orakɯ] remain unexplained. Turbidity resembles the present analysis in the sense that the distinction between segments is encoded in their representations, but differs from it by encoding it in output rather than input representations. However, as Baković (2007: 230) argues, although Velar Deletion can also make use of the projection/pronunciation distinction, it does not have to, and such a distinction is only motivated to account for SDF.

5.2 NDEB in Parallel OT

Classic treatments of NDEB include local constraint conjunction (Łubowicz 2002) and comparative markedness (McCarthy 2003). Local constraint conjunction combines two constraints and penalises candidates that violate both of them in a local domain (Łubowicz 2002: 247). By conjoining a markedness constraint with R-ANCHOR(Stem;σ), which requires the stem-final segment in the input to align with the right edge of a syllable in the output, Łubowicz (2002: §4.2) captured velar palatalisation that only occurs in morphologically derived environments in Polish. This works for Velar Deletion and h-deletion here: *VkV & R-ANCHOR(Stem;σ) $\gg$ MAX penalises *[be.be.ki], in which the stem-final /k/ is resyllabified as an onset, but not [a.vu.kat] because the velar is not stem-final, or [se.ne.ki] because the stem-final [e] remains at the right edge of a syllable. However, local conjunction always preserves suffix-initial consonants, so it cannot account for Elision, n-deletion or Continuant Deletion: a consonant-initial suffix does not resyllabify the stem-final consonant. For example, *CC & R-ANCHOR(Stem;σ) does not penalise *[tʃan.sɯ] but [tʃa.nɯ] instead, because the latter misaligns the stem edge.

Comparative markedness splits each markedness constraint M into a 'new' ($_{\text{N}}$M) and an 'old' ($_{\text{O}}$M) version. Every candidate is compared with the fully-faithful candidate: old markedness violations are incurred by structures already in it, and new ones are introduced by unfaithful mappings (McCarthy 2003: 1–2). For morphologically derived environments, McCarthy compares each candidate with its base through output–output

(OO) correspondence (Benua 1997), and a new OO comparative markedness constraint is violated if the violating segment has a correspondent in the base that is in a different environment, or if it does not have a correspondent at all. For example, when OO-$_{N}$*VkV ≫ MAX, the velar of /bebek+i/ is deleted, because its correspondent in the base [bebek] is not intervocalic. But the velar of [avukat] is preserved, because [avukat] has no simpler base and has only old violations. However, the comparison with the base fails on suffix-initial segments that do not delete: without correspondents in the base, suffix-initial segments in a marked environment always incur new violations and are deleted. This prediction would be wrong for *-ki* ([seneki]) and many other non-alternating consonant-initial suffixes, such as *-sIz* ([duraksɯz]) and *-lAr* ([bebekler]).[21]

Other accounts use constraints protecting morpheme-internal sequences, e.g. I-CONTIG (McCarthy & Prince 1995), NEIGHBOURHOOD^2 (Itô & Mester 1996; Inkelas 2000: §2.1), and Sequential Faithfulness constraints (Burzio 2000). I-CONTIG and NEIGHBOURHOOD^2 both favour output strings whose corresponding input is a contiguous string. Sequential Faithfulness constraints forbid any change to a specified input sequence in the UR of a single morpheme. In this way, morpheme-internal changes are penalised because they destroy string contiguity or sequences in the UR with a more privileged status of faithfulness. However, deleting a segment at a morpheme edge does not break the contiguity of the morpheme, so I-CONTIG, for example, protects neither the stem-final velar of /orak/ nor the suffix-initial consonants of *-ki* and *-sIz*.

The most straightforward way of restricting a process to apply only in morphologically derived environments is to use boundary-referring constraints, although it is not a general solution to NDEB. For example, a constraint against a stem-final velar between vowels (*Vk+V) penalises *[bebeki] but not [avukat], so morpheme-internal velars are protected. Such constraints can also account for the asymmetry between morpheme-final and morpheme-initial velars: *Vk+V penalises only the former (*[bebeki]) but not the latter ([seneki]). However, stem-final velars in lexical exceptions like /orak/ would be incorrectly deleted.

All four accounts thus capture the edge-only nature of NDEB for some rules, but each treats segments at the same position in the same manner, whereas alternation varies from morpheme to morpheme for the processes under discussion. It is not the case that segments at the same position are treated in the same way, yet these analyses predict so. So, lexically encoding this contrast between segments might be the best way to accurately capture all and only the alternating segments.

Despite their ability to at least describe NDEB, none of the aforementioned mechanisms accounts for SDF at the same time. For a grammar to select [ajaɯ] over [ajasɯ] on the local conjunction analysis, a constraint conjoining MAX and markedness constraints like *VsV must outrank MAX, which in turn dominates *VsV. In this way, [ajaɯ] wins because it does not violate *VsV, which [ajasɯ] does. On the comparative markedness analysis, [ajaɯ] only wins if a constraint penalising new marked structures created by the mapping /ajak+sɯ/ → [ajasɯ] (e.g. IO-$_{N}$*VsV) dominates MAX, which in turn outranks those targeting old markedness violations. But the penalty of such a surface structure is questionable: *VsV is only motivated by the SDF output and intervocalic /s/ is in fact pervasive in Turkish.

Privileging contiguous strings within a single morpheme only distinguishes a candidate making morpheme-internal changes from one making changes at morpheme edges.

[21] The definition of the base might also pose problems for Japanese bound verb roots like /tob-/: they do not have free-standing forms, so the affixed forms have only old violations, and Continuant Deletion cannot be derived.

However, both the SDF output and the transparent candidate only make changes at morpheme boundaries. For example, in /ajak+sɯ/, the transparent [ajasɯ] deletes nothing morpheme-internally, and it differs from the SDF output [ajaɯ] only by one suffix-initial segment. A faithfulness constraint like I-Contig or Neighbourhood^2 assigns [ajaɯ] at least as many violations as [ajasɯ]. A boundary-referring constraint targeting the only differing segment /s/ (e.g. *V+sV) would have no independent motivation and make wrong predictions for forms like [ʃiʃesi]. So, the SDF output still loses. To derive SDF, these analyses would need one of the mechanisms discussed in Section 5.1. In comparison, the present account captures NDEB and SDF with the same representational distinction.

6 Discussion

6.1 Richness of the Base and the lexicon

An underspecification account of NDEB raises the question of how it relates to Richness of the Base (ROTB), which Rasin (2025) rejected because it allows too many underlying representations for theories of NDEB. When the underspecification account is combined with restrictions on the lexical distribution of underspecified segments, the puzzles of persistent blocking, blocking within suffixes, and non-contrastive triggers, which most competing NDEB theories fail to explain, can be properly solved (Rasin 2025: §4).[22] In his account, the lexical restrictions are represented as language-specific morpheme structure constraints, in the same format as ordinary rules. They apply to morphemes in the lexicon and determine which segments are underspecified. Ordinary rules then apply to derive SFs.

The present account does not need morpheme structure constraints, and it is compatible with ROTB. Any segment at any position in a morpheme can be either under- or fully-specified, and Lexicon Optimisation selects among the possible inputs (Inkelas 1995). This is because the processes differ from those discussed by Rasin (2025) in two ways. First, the alternations here are lexically contrastive, whereas those used by Rasin (2025) to reject ROTB are obligatory in derived environments. Because both alternating and non-alternating versions of a stem-final velar or a suffix-initial /s/ are attested ([bebek] vs. [orak]; *-(s)I* vs. *-sIz*), such a contrast is needed in the UR and ROTB generates it. Second, the positional restrictions can follow from the grammar without restrictions on input (Section 4.7). Underspecified segments in positions where they could not alternate are indistinguishable from either absence or association, so no input restriction is needed to exclude them. The language-general process of Japanese w-deletion can also be derived by adding the ranking *w[–low] ≫ MaxLink, which the data permit, rather than by a restriction on inputs.

So, for a process that is obligatory in derived environments and blocked in nonderived ones, morpheme structure constraints would fare better because ROTB may allow fully-specified input segments to appear in a position where they must be underspecified. If so, the rule applies only optionally. Should morpheme structure constraints prove necessary elsewhere, the present analysis would be unaffected, because such rules would restrict the underspecified inputs to those that Lexicon Optimisation selects anyway.

[22] Rasin (2025: 739) acknowledges that comparative markedness (McCarthy 2003) "succeeds on all three puzzles", but leaves it out of the discussion because it has been argued against in the literature and he has nothing further to contribute to the evaluation.

6.2 Contextual faithfulness, expressivity and factorial typology

An anonymous reviewer points out that contextual faithfulness may be overly expressive. McCarthy (2007b: 25–27) rejects the usage of contextual faithfulness as a general solution to counterfeeding in Parallel OT, since "[d]ealing with the full range of counterfeeding interactions will require a very rich faithfulness theory, undoubtedly much richer than we want or would otherwise need". Hauser & Hughto (2020: 4, 19, 21–22) respond that the analytical potential of such constraints outweighs the risk of overgeneration: an overly rich faithfulness theory need not lead to pathological typologies, and contextual faithfulness constraints could instead be induced on a language-specific basis, so that they never enter the factorial typology.

Following Hauser & Hughto (2020: 19–21), I computed the factorial typology with and without the contextual faithfulness constraints over ten Turkish inputs: /bebeK/, /bebeK+ler/, /bebeK+i/, /ʧan+Sɯ/, /ʃiʃe+Si/, /ip+In/, /anne+In/, /avukat/, /ajaK+Sɯ/ and /bebeK+In/. For each input, the candidates are all assignments of one of three statuses (association, non-association, deletion) to each melodic unit. So, an input of n units has 3^n candidates. All total orders (40,320 with the contextual constraints, 720 without) were evaluated over this candidate space with the script used for the ranking checks (Appendix B). The typology was replicated with OT-Help 2.0 (Staubs et al. 2010) on the same candidate sets. The two typologies are compared to see which languages arise from the contextual constraints.

Without MAX-C/V_V and MAX-V/C_C, these inputs yield only eight languages (five if only unique optima are counted). In each of them, the underspecified suffixal segments in /ʧan+Sɯ/, /ʃiʃe+Si/, /ip+In/ and /anne+In/ surface either in all contexts or in none. In other words, there is neither the consonant∼∅ alternation of [ʃiʃesi] and [ʧanɯ] nor the vowel∼∅ alternation of [ipin] and [annen]. Consequently, there is no SDF. [bebein] is never generated, and [ajaɯ] occurs only in languages in which the suffix-initial /S/ never surfaces, which means [ajaɯ] does not arise as a result of Elision.

With the contextual constraints, 81 distinct languages are possible,[23] including the attested Turkish one and the aforementioned eight. The properties of these languages are summarised in (22). Among these languages, 45 keep an underspecified suffix segment only where its input neighbours belong to the opposite class. That is, a consonant between vowels or a vowel between consonants.

This set of constraints ensures that the transparent candidate never wins in the consonantal cases (22a), and that SDF naturally falls out in languages with Velar Deletion and Elision (22b) as attested in Turkish. A language that deletes an underspecified preconsonantal stem consonant while keeping an underspecified postconsonantal suffix consonant in a C_1C_2 sequence at a morpheme juncture cannot be generated with this constraint set (22c).

(22) Properties of the factorial typology

a. *The transparent output*: For consonant-SDF, the transparent candidate [ajasɯ] never wins; only [ajaɯ] (in 64 languages), [ajaksɯ] (in 13 languages) and [ajakɯ] (in 4 languages) occur. For vowel-SDF, [beben] is only found in languages without vowel ∼∅ alternations.

[23] Languages are individuated by their pronounced outputs, and are allowed to have multiple equally optimal outputs for an input. The input with tied candidates is /avukat/ in all the 26 languages containing ties. If only languages with a unique optimum for every input are allowed, as in OT-Help 2.0 (Staubs et al. 2010), the total number drops to 55. The properties in (22) hold under both counts.

b. *Entailment of SDF*: For consonant-SDF, every language in which an underspecified stem-final velar is lost between vowels (i.e. with Velar Deletion) and an underspecified suffix consonant is lost after a consonant (i.e. with Elision) maps /ajaK+Sɯ/ to [ajaɯ]. In other words, consonant-SDF is entailed in languages with these simple alternations. The counterfeeding output [ajakɯ] occurs only in languages in which underspecified velars survive before vowels ([bebeki]). For vowel-SDF, only an additional ranking MAX-V/C_C ≫ MAX-C/V_V is needed.

c. *Stems are at least as faithful as suffixes.* No language loses an underspecified stem segment while keeping an underspecified suffix segment in a comparable context. For instance, /bebeK+ler/ → [bebeler] never co-occurs with /ʧan+Sɯ/ → [ʧansɯ], because violating DEPLINK$_{\text{suffix}}$ is at least as costly as violating DEPLINK.

In brief, as in Hauser & Hughto (2020: 20), the contextual constraints add the context-sensitive alternations that the analysis wants to capture, without pathological typological predictions. Without the contextual faithfulness constraints, SDF cannot be captured.

6.3 Scope

Although the present analysis accounts for the four SDF cases identified in Section 2, the cases examined here are not necessarily an exhaustive list of all possible SDF cases, and the current analysis may not extend to other cases of SDF that might exist. An anonymous reviewer suggests a hypothetical SDF interaction involving featural changes that also fits Baković's (2011) definition. As shown in (23a), Dissimilation can self-destructively feed Epenthesis, because two dissimilar consonants create the environment for Epenthesis, and the epenthetic vowel makes the two consonants no longer adjacent and the application of Dissimilation is unjustified.

(23) Hypothetical SDF with a feature-changing rule

	UR	a. /əss/	b. /əssə/	c. /əsz/
Dissimilation:	s → z / s_	z	z	
Epenthesis:	∅ → i / s_z#	i		i
	SF	[əsiz]	[əszə]	[əsiz]

The present analysis does not extend to this case, because it only controls the realisation or non-realisation of underspecified segments. The [z] in Dissimilation does not alternate with ∅. Rule ordering, on the other hand, can account for SDF between two rules with such featural changes.

In other words, the present account has reduced all *reported* SDF cases to an epiphenomenon of the realisation of lexically designated underspecified segments at morpheme junctures, rather than an interaction the grammar has to encode separately, but other possible SDF cases (e.g. one that involves feature-changing processes) would still require rule ordering or other mechanisms. Should such cases exist, they might also pose challenges to Parallel OT. I leave the discovery of such cases to future research.

7 Conclusion

This paper has re-examined four cases of SDF, two in Turkish, one in Javanese and one in Japanese. In all four, at least one of the processes involved is confined to morpheme junctures and sometimes restricted to particular morphemes. They also avoid stem–suffix consonant sequences in a similar way, by realising a suffix-initial segment as required by the context. I have proposed that the alternating segments are underspecified: present in the input but unassociated to the skeletal tier. With constraints requiring faithfulness to association, a higher cost for associating suffix segments, and two faithfulness constraints with input-defined contexts, Parallel OT derives the simple alternations and their restriction to junctures, and more importantly, the SDF outputs in a single parallel evaluation.

This analysis has three implications. First, these results show that SDF involving alternations with ∅ does not have to be accounted for by serial derivations and does not pose challenges to Parallel OT once the underlying representations of segments are properly understood. Second, SDF naturally falls out from the representation of the alternating segments and constraint rankings responsible for simple alternations in each language. Moreover, the typology of the current constraint set also shows that consonant-SDF is obligatory whenever the two simple alternations co-occur. Vowel-SDF needs only one additional ranking. Finally, in contrast with some previous analyses, the restriction of alternations to certain positions follows from the grammar rather than from restrictions on inputs, making the analysis compatible with essential premises of Parallel OT such as ROTB.

A Javanese and Japanese tableaux

The tableaux below use the constraint definitions in (12), with *VhV for Javanese and *w[–low] for Japanese in place of *VkV. MAX-V/C_C is not shown in the Javanese tableaux, since no Javanese ranking depends on it. It is included in the Japanese tableaux, because it is needed for the alternations of the underspecified initial vowel of the negative suffix (29) (cf. Section 4.6).

A.1 Javanese

In /omaH+Ne/, neither underspecified segment is protected by MAX-C/V_V. Realising /N/ violates $\text{DEPLINK}_{\text{suffix}}$, and realising only /H/ creates an intervocalic [h], so both are removed (24). (25) shows that /N/ is realised after a vowel-final stem but not after a consonant-final one. (26) shows that underspecified /H/ is lost between vowels but realised word-finally, and fully-specified /h/ is preserved between vowels.

(24) Javanese SDF

/omaH+Ne/	SPECIFY	MAXLINK	*VhV	MAX-C/V_V	$\text{DEPLINK}_{\text{suffix}}$	MAX	DEPLINK
a. ☞ omae						**	
b. omane					*!	*	*
c. omahne					*!		**
d. omahe			*!			*	*
e. omaHNe	*!*						

(25) The underspecified suffix consonant after vowel- and consonant-final stems

a.

/kopi+Ne/	SPECIFY	MAXLINK	*VhV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ kopine					*		*
b. kopie				*!		*	
c. kopiNe	*!						

b.

/kuliṯ+Ne/	SPECIFY	MAXLINK	*VhV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ kuliṯe						*	
b. kuliṯne					*!		*
c. kuline		*!			*	*	*
d. kulie		*!				**	
e. kuliṯNe	*!						

(26) Underspecified stem-final /H/ and fully-specified /h/

a.

/səkolaH+an/	SPECIFY	MAXLINK	*VhV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ səkolaan				*		*	
b. səkolahan			*!				*
c. səkolaHan	*!						

b.

/omaH/	SPECIFY	MAXLINK	*VhV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ omah							*
b. oma						*!	
c. omaH	*!						

c.

/dihin/	SPECIFY	MAXLINK	*VhV	MAX-C/V_V	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ dihin			*				
b. diin		*!		*		*	
c. diHin	*!	*					

A.2 Japanese

In the SDF input /kaW+Ru/, neither underspecified consonant is protected by MAX-C/V_V, so they are both deleted, as in Turkish and Javanese (27). (28) shows that suffix-initial /R/ is preserved after vowels but deleted after consonants, and fully-specified /r/ is retained after a consonant. Similarly, (29) shows that suffix-initial vowels are kept after consonants but removed after vowels. (30) shows that underspecified /W/ surfaces

before [a] because it does not violate *w[–low]; MAX ≫ DEPLINK then keeps /W/. In [kawase], /S/ does not surface because $\text{DEPLINK}_{\text{suffix}}$ ≫ MAX. Finally, /iW+e+Ru/ and /kaW+e/ show that /W/ is lost before a non-low vowel even when it is protected by its input context, rather than being saved by deleting the following vowel (31).

(27) Japanese SDF

/kaW+Ru/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	$\text{DEPLINK}_{\text{suffix}}$	MAX	DEPLINK
a. ☞ kau							**	
b. karu						*!	*	*
c. kawru						*!		**
d. kawu		*!					*	*
e. kaWRu	*!*							

(28) Underspecified suffix-initial /R/ and fully-specified /r/

a.

/tob+Ru/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	$\text{DEPLINK}_{\text{suffix}}$	MAX	DEPLINK
a. ☞ tobu							*	
b. tobru						*!		*
c. toru			*!			*	*	*
d. tou			*!				**	
e. tobRu	*!							

b.

/tabe+Ru/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	$\text{DEPLINK}_{\text{suffix}}$	MAX	DEPLINK
a. ☞ taberu						*		*
b. tabeu				*!			*	
c. tabeRu	*!							

c.

/nenrei/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	$\text{DEPLINK}_{\text{suffix}}$	MAX	DEPLINK
a. ☞ nenrei								
b. nenei			*!				*	
c. neNrei	*!		*					

(29) The underspecified suffix-initial vowel after consonant- and vowel-final stems

a.

/tob+Anai/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	$\text{DEPLINK}_{\text{suffix}}$	MAX	DEPLINK
a. ☞ tobanai						*		*
b. tobnai					*!		*	
c. tobAnai	*!							

b.

/tabe+Anai/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ tabenai							*	
b. tabeanai						*!		*
c. tabeAnai	*!							

(30) Underspecified stem-final /W/ before [a]

a.

/kaW+Sase/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ kawase							*	*
b. kaase							**!	
c. kasase						*!	*	*
d. kawsase						*!		**
e. kaWSase	*!*							

b.

/kaW+Anai/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ kawanai						*		**
b. kawnai					*!		*	*
c. kaanai				*!		*	*	*
d. kaWanai	*!					*		*
e. kawAnai	*!							*

(31) Underspecified /W/ before a non-low vowel

a.

/iW+e+Ru/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ ieru				*		*	*	*
b. iweru		*!				*		**
c. iwru			*!		*	*	*	**
d. iWeru	*!					*		*

b.

/kaW+e/	SPECIFY	*w[–low]	MAXLINK	MAX-C/V_V	MAX-V/C_C	DEPLINK$_{suffix}$	MAX	DEPLINK
a. ☞ kae				*			*	
b. kawe		*!						*
c. kaw			*!				*	*
d. kaWe	*!							

B Computational verification

All rankings and tableaux in this paper were checked with a Python 3 script written with the assistance of Claude Opus 5.5. Each constraint in (12) is implemented as a function counting the violations of a candidate. The candidate space for each input is described in Section 6.2. The attested output was compared with every other candidate, yielding a set of elementary ranking conditions (Prince 2002), whose consistency was tested by Recursive Constraint Demotion (Tesar & Smolensky 2000). A ranking A ≫ B was taken to be necessary if adding the condition B ≫ A makes the set inconsistent. The rankings were replicated with OT-Help 2.0 (Staubs et al. 2010) on the same candidate sets.

The comparative tableau in Table 16 summarises the Turkish ranking arguments of Sections 4.4 and 4.5 (Prince 2002).[24] Each row compares the attested output with one competitor; W marks a constraint that favours the winner and L one that favours the loser. Rows a–g correspond to the simple alternations, row h to vowel-SDF and rows i–j to consonant-SDF. Rows i and j introduce no new ranking.

Input: winner ~ loser	Spec	MaxV	MLink	*VkV	MaxC	DL_{suf}	Max	DL
a. /bebeK/: [bebek] ~ [bebe]							W	L
b. /ʧan+Sɯ/: [ʧanɯ] ~ [ʧansɯ]						W	L	W
c. /ʃiʃe+Si/: [ʃiʃesi] ~ [ʃiʃei]					W	L	W	L
d. /ip+In/: [ipin] ~ [ipn]		W				L	W	L
e. /bebeK+i/: [bebei] ~ [bebeki]				W	L		L	W
f. /bebeK+i/: [bebei] ~ [bebeKi]	W				L		L	
g. /avukat/: [avukat] ~ [avuat]			W	L	W		W	
h. /bebeK+In/: [bebein] ~ [bebekn]		W			L	L		
i. /ajaK+Sɯ/: [ajaɯ] ~ [ajasɯ]						W	L	W
j. /ajaK+Sɯ/: [ajaɯ] ~ [ajakɯ]				W			L	W

Table 16: Comparative tableau of the Turkish ranking arguments. Suffixes: *-i* = Acc *-(y)I*; *-Sɯ* and *-Si* = 3sg.Poss *-(s)I*; *-In* = 2sg.Poss *-(I)n*

The results of the verification are summarised in (32).

(32) Results of the verification

a. *Turkish* (13 inputs: those in (13)–(19) and /orak+ɯ/): consistent. Every competitor shown in the tableaux loses, and no other candidate in the full candidate space wins. The necessary rankings are those in Figure 1, and 12 of the 40,320 total orders of the eight constraints are compatible.

b. *Turkish, alternative constraint sets*: without MaxLink, inconsistent;
with Max_{full} (which penalises deletion of fully-specified segments) instead of MaxLink, consistent, but Specify ≫ *VkV is then additionally required to exclude delinked candidates such as [avuKat];
with Max_{full} and Anchor (penalising the loss of the stem-final segment) added, consistent, but Anchor is not ranked with respect to Max and does no work;
without $DepLink_{suffix}$, Max-C/V_V or Max-V/C_C, inconsistent;

[24] Constraint names are abbreviated: Spec = Specify, MaxV = Max-V/C_C, MLink = MaxLink, MaxC = Max-C/V_V, DL_{suf} = $DepLink_{suffix}$ and DL = DepLink.

with ONSET in place of MAX-C/V_V, or *CC# in place of MAX-V/C_C, inconsistent.

c. *Entailment*: Turkish: the simple alternations alone (i.e. all inputs except /ajaK+Sɯ/ and /bebeK+In/) are compatible with 15 total orders; all of them select [ajaɯ]. Twelve of them select [bebein], and three select [bebekn].[25]
Javanese: the simple alternations are compatible with 24 orders, all of which select [omae].
Japanese: the simple alternations are compatible with 30 orders, all of which select [kau].

d. *Javanese* (6 inputs) and *Japanese* (10 inputs), with the full constraint set in (12): consistent, with the rankings in (20); 24 and 30 of the 40,320 total orders are compatible, respectively.

e. *Factorial typology*: see Section 6.2.

Abbreviations

ACC = accusative, CAUS = causative, COND = conditional, DEF = definite, IMP = imperative, NEG = negative, NPST = nonpast, PL = plural, POSS = possessive, PRED = predicative, SG = singular, VOL = volitional, 1/2/3 = first/second/third person; NDEB = nonderived environment blocking, OT = Optimality Theory, SDF = self-destructive feeding, SF = surface form, UR = underlying representation

Competing interests

The author has no competing interests to declare.

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[25] Only vowel-SDF establishes that MAX-V/C_C ≫ MAX-C/V_V (Section 4.5), so the simple alternations permit MAX-C/V_V ≫ MAX-V/C_C. These three rankings are exactly those where MAX-C/V_V ≫ MAX-V/C_C.

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